



Institutional Loan Syndication & NPL Marketplace

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Project: Blockchain-Based Asset Management - Proof of Concept (PoC) This Proof of Concept (PoC) demonstrates a 2026-standard financial infrastructure for the lifecycle management of multifamily loans. It uses ERC-1155 Smart Contracts to manage both the unique underlying asset (NFT) and its fractional participation shares (Fungible Tokens).



Key Features

- **On-Chain Digital Twin:** Every loan is minted as a unique digital asset with a verified contract address.
- **Atomic Syndication:** Lead banks can distribute risk to consortium partners with T+0 instant settlement.
- **NPL Pivot Logic:** Real-time transition from "Performing" to "Non-Performing" status, triggering secondary market listing and valuation haircuts.
- **Dual-Role Dashboard:** Toggle between Lead Bank (Seller) and Hedge Fund (Buyer) perspectives.
- **Digital Vault:** Integrated access to IPFS-verified Appraisal Reports and Phase I Environmental documents.



Technical Stack

- **Blockchain Engine:** Hardhat (Local Private Ethereum Node)
- **Smart Contracts:** Solidity (OpenZeppelin ERC-1155 Standard)
- **Frontend/UI:** Streamlit (Python)
- **Web3 Bridge:** Web3.py
- **Environment:** GitHub Codespaces

Setup & Installation

1. Initialize Environment Open the terminal in GitHub Codespaces and install the necessary dependencies:

```
npm install @openzeppelin/contracts
pip install streamlit web3
```

2. Start the Private Blockchain In a dedicated terminal, launch the local Ethereum node:

```
npx hardhat node
```

Keep this terminal open to maintain the network.

3. Deploy the Smart Contract In a second terminal, deploy the LoanSyndication.sol contract:

```
npx hardhat ignition deploy ./ignition/modules/LoanSyndication.js --network localhost
```

Copy the Contract Address (e.g., 0x5Fb...) and ensure it matches the CONTRACT_ADDRESS in app.py.

4. Launch the Dashboard In a third terminal, start the Streamlit application:

```
python3 -m streamlit run app.py
```

Demo Workflow

- 1. Syndication:** As a Lead Bank, select a partner from the dropdown and syndicate \$1,000,000. Refresh to see the "On-Chain Funding" metric update.
- 2. Distress Simulation:** Use the sidebar slider to move the loan to Non-Performing (NPL). Observe the valuation "Haircut" chart.

3. **Secondary Trade:** Switch roles to Hedge Fund, review the Appraisal in the Digital Vault, and purchase the distressed debt at the listed market discount.

User Guide: Institutional NPL & Syndication Marketplace

Framework: Ethereum (ERC-1155) & Streamlit

1. **Overview** This platform provides a blockchain-based "Digital Twin" environment for managing multifamily loans. It enables Lead Banks to syndicate primary debt and manage distressed assets (NPLs) via a transparent, on-chain secondary marketplace.

2. System Roles

Lead Bank (Seller)

- **Primary Action:** Distribute loan risk to partner institutions.
- **NPL Management:** Toggle asset status to "Non-Performing" to trigger secondary market listings and capital haircuts.
- **Vault Control:** Manage the "Digital Vault" containing appraisals and environmental reports.

Hedge Fund (Buyer)

- **Primary Action:** Scout for distressed debt opportunities.
- **Due Diligence:** Review on-chain verified documents in the Digital Vault.
- **Execution:** Purchase fractional loan shares at a market-determined discount.

3. Operational Workflow

Phase A: Primary Syndication

1. Ensure the status is set to "Performing."
2. Select a Partner Bank address from the consortium dropdown.
3. Enter the Syndication Amount (e.g., \$1,000,000).
4. Click "Finalize Syndication" to mint shares directly to the partner's wallet.

Phase B: NPL Pivot & Valuation

1. Toggle the Loan Performance Status to "Non-Performing (NPL)."
2. The dashboard will shift to a "Distressed" state (Red UI).
3. Use the Secondary Market Slider to set the "Ask Price" (e.g., 65% of Par).
4. Review the Valuation Analysis chart to visualize the implied capital haircut.

Phase C: Secondary Sale

1. Click "List NPL for Secondary Sale" to broadcast the asset.
2. Switch the view to "Hedge Fund (Buyer)."
3. Execute the purchase to simulate the transfer of distressed debt to a third-party fund.

4. The Digital Vault

Located in the sidebar, the Digital Vault provides an immutable link between the financial asset and its physical due diligence.

- **Appraisal Reports:** Real-time property valuations.
- **Phase I Environmental:** Risk assessment documentation.
- **Verification:** All documents are noted as "Verified On-Chain," representing an IPFS-backed audit trail.

5. Technical Recovery (Next Session)

To restart the environment tomorrow, run these commands in order:

1. `npx hardhat node`
2. `npx hardhat ignition deploy ./ignition/modules/LoanSyndication.js --network localhost`

Troubleshooting & Maintenance

1. "Blockchain Not Found" Error

- **Cause:** The Hardhat node is not running or has timed out.
- **Solution:** Go to your first terminal and ensure `npx hardhat node` is active. If the terminal is empty, re-run the command.

2. "Invalid EIP-55 Checksum"

- **Cause:** The `CONTRACT_ADDRESS` in `app.py` has a capitalization error.
- **Solution:** Ensure the address is wrapped in `w3.to_checksum_address()`.
- **Code Check:** `CONTRACT_ADDRESS = w3.to_checksum_address("0x...")`

3. "Total Funded" is not updating

- **Cause:** Streamlit has not refreshed its connection to the blockchain.
- **Solution:** Click the three dots (:) in the top-right of the dashboard and select Rerun, or refresh your browser tab.

4. "Address has no balance" (Transaction Fails)

- **Cause:** You are trying to syndicate a loan, but the Hardhat node was restarted and the contract address changed.
- **Solution:** Re-deploy the contract using the `ignition deploy` command and update the `CONTRACT_ADDRESS` in `app.py` with the new address provided by the terminal.

5. Environment Reset (The "Clean Slate" Fix)

If the app is behaving unexpectedly, perform a full reset:

1. **Stop everything:** Press `Ctrl + C` in all active terminals.
2. **Clear Cache:** Run `npx hardhat clean`.
3. **Restart Node:** Run `npx hardhat node`.
4. **Re-deploy:** Run your `ignition deploy` command.
5. **Re-launch App:** Run `python3 -m streamlit run app.py`.



Screenshots

User Identity

View Dashboard As:

Lead Bank (Seller)

Digital Vault

Secured via IPFS Hash Verification

> View Asset Documents

Asset Controls

Loan Performance Status

Performing

User Identity

View Dashboard As:

Lead Bank (Seller)

Digital Vault

Secured via IPFS Hash Verification

> View Asset Documents

Asset Controls

Loan Performance Status

Delinquent

PerformingNon-Performing (NPL)

Loan Marketplace: Performing

Asset Digital Twin: 0x5Fb0B2315678afecb367f032d93F642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

6.25%

Lead Bank Management Console

Loan is healthy. Primary syndication active.

Select Institution:

0x70997970C51812dc3A010C7d01b50e0d17dc79C8

Syndication Amount (\$)

1000000

Finalize Syndication

Lender Role	Total Exposure
Lead Bank	\$10,000,000 (Gross)
Active Partners/Buyers	\$0 (Net)

Loan Marketplace: Delinquent

Asset Digital Twin: 0x5Fb0B2315678afecb367f032d93F642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

45.0% Discount

Lead Bank Management Console



User Identity

View Dashboard As:

Lead Bank (Seller)

Digital Vault

Secured via IPFS Hash Verification

View Asset Documents

Validated On-Chain: ✔

Appraisal_Report_2026.pdf

Phase_I_Environmental.pdf

Property_Condition_Assess.pdf

Vault Access Logged to Blockchain

User Identity

View Dashboard As:

Hedge Fund (Buyer)

Digital Vault

Secured via IPFS Hash Verification

View Asset Documents

Asset Controls

Loan Performance Status

Non-Performing (NPL)

Loan Marketplace: Non-Performing (NPL)

Asset Digital Twin: 0x5FbDB2315678afecb367f032d93f642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

45.0% Discount

Lead Bank Management Console



Loan Marketplace: Non-Performing (NPL)

Asset Digital Twin: 0x5FbDB2315678afecb367f032d93f642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

45.0% Discount

Institutional Buyer Portal

OPPORTUNITY: Distressed Loan available at 60% of Par

Investment Amount (\$)

2000000

Par Value of Shares

\$2,000,000

Acquisition Cost

\$1,200,000

↓ -40%

Confirm Secondary Purchase

Lender Role	Total Exposure
Lead Bank	\$10,000,000 (Gross)
Active Partners/Buyers	\$0 (Net)

User Identity

View Dashboard As:

Hedge Fund (Buyer)

Digital Vault

Secured via IPFS Hash Verification

> View Asset Documents

Asset Controls

Loan Performance Status

Delinquent

Performing

Non-Performing (NPL)

User Identity

View Dashboard As:

Lead Bank (Seller)

Digital Vault

Secured via IPFS Hash Verification

> View Asset Documents

Asset Controls

Loan Performance Status

Performing

Loan Marketplace: Delinquent

Asset Digital Twin: 0x5FbD82315678afech367f032d93f642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

45.0% Discount

Institutional Buyer Portal

No distressed opportunities available at this time.

Lender Role	Total Exposure
Lead Bank	\$10,000,000 (Gross)
Active Partners/Buyers	\$0 (Net)

Asset Digital Twin: 0x5FbD82315678afech367f032d93f642f64180aa3

Principal Balance

\$10,000,000

On-Chain Syndication

\$0

Yield / Pricing

6.25%

Lead Bank Management Console

Loan is healthy. Primary syndication active.

Select Institution:

0x70997970C51812dc3A010C7d01b50e0d17dc79C8

0x70997970C51812dc3A010C7d01b50e0d17dc79C8

0x3C44CdDb6a900fa2b585dd299e03d12FA4293BC

0x90F79bf6EB2c4f870365E785982E1f101E93b906

0x15d3AAf54267DB7D7c367839AA71A00a2C6A65

Lender Role	Total Exposure
Lead Bank	\$10,000,000 (Gross)
Active Partners/Buyers	\$0 (Net)